

Notes to paperes

Some subtitle for my thesis

Johan Ulstrup

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1 index

1.1 evolution on X

The production of gametes in sexually reproducing organisms is a complex, tightly regulated process involving numerous cellular and molecular mechanisms. In males, spermatogenesis—the process by which sperm are formed—progresses through four main stages of differentiation: from germ cells to spermatogonia, then to pachytene spermatocytes, round spermatids, and finally to mature spermatozoa (Wang et al. 2019). This process is fundamental to male fertility. Meiosis, a specialized form of cell division, ensures the proper pairing, recombination, and segregation of homologous chromosomes, thereby maintaining genetic integrity and promoting variation. A deep understanding of these molecular events is essential for comprehending inheritance, speciation, population genetics, and the biological causes of infertility.

Sex chromosomes differ from autosomes in several key ways due to their distinct patterns of inheritance and copy number. The Y chromosome is found only in males and exists as a single copy. In contrast, the X chromosome follows a more complex inheritance pattern: females carry two copies, while males carry only one. Consequently, the X chromosome resides two-thirds of the time in females and one-third in males, a distribution that influences how selection acts upon it. In males, the hemizygous nature of the X chromosome means that any mutations or deleterious alleles are fully exposed, without a second copy to mask their effects.

This feature is central to Haldane's rule (Haldane 1922), which posits that in hybrids between two species, the heterogametic sex (e.g., XY in mammals or ZW in birds) is more likely to be inviable or sterile. While widely observed, the precise causes of this pattern remain a subject of ongoing debate. Several hypotheses have been proposed to explain it. One suggests that incompatibilities involving the Y chromosome can arise if it fails to remain compatible with the X chromosome or autosomes. Another centers on dosage compensation, proposing that hybridization may disrupt the regulatory mechanisms that balance gene expression between sex chromosomes. The dominance hypothesis argues that recessive deleterious alleles may be unmasked in the heterogametic sex, leading to sterility. Additional ideas include the faster evolution of male-biased reproductive genes, which can result in functional mismatches, and the accelerated divergence of X-linked loci compared to autosomes, known as faster-X evolution. Lastly, meiotic drive—genetic conflicts between selfish elements on sex chromosomes—has also been implicated in hybrid sterility.

These hypotheses underscore the complex interplay between sex chromosomes, selection, and hybridization (Cowell 2023). Although Haldane's rule is consistently observed across diverse

taxa—and even across kingdoms—the underlying mechanisms are not universally agreed upon. In many cases, multiple processes may act together ? (Lindholm et al. 2016).

The X chromosome, in particular, appears to be subject to strong and unique selective pressures, especially in primates. This topic is explored in greater depth in the following sections.

1.2 Chromatin structure

Regions under strong selection across primate species—including humans and baboons—are consistently found on the X chromosome, spanning megabase-scale genomic areas. As previously noted, factors such as chromosomal architecture and structural rearrangements may play crucial roles in enabling, regulating, or insulating these regions from selective pressures.

Understanding genome organization and variation is therefore essential for uncovering the mechanisms behind selection. Chromatin has long been recognized as central to gene regulation and cellular function (Lieberman-Aiden et al. 2009), in part because the 3D structure of chromosomes brings distant genomic elements into close contact. Disruption of this spatial organization can lead to developmental abnormalities ? (Dixon et al. 2015). Chromatin is organized hierarchically, with multiple levels of structure nested within one another (see Figure 2). At the broadest level, chromatin compartments can span several megabases ? (Lieberman-Aiden et al. 2009), while at finer scales, even 500 base pairs can contribute to subgenomic structural organization. Structures such as topologically associating domains (TADs) and chromatin loops, which operate at sub-megabase scales, further help partition regulatory elements and maintain proper genomic function ??(Ramírez et al. 2018; Zuo et al. 2021).

Chromatin, the complex of DNA wrapped around histone proteins and associated with various regulatory factors, plays a central role in the spatial and temporal control of gene expression, as well as in maintaining the structural integrity and stability of the genome. Far from being a static scaffold, chromatin is a highly dynamic entity, subject to extensive remodeling in response to cellular signals, developmental cues, and environmental stimuli. Its organization within the nucleus is hierarchical, ranging from nucleosomes to higher-order domains, and this architecture can differ substantially across cell types, developmental stages, and even between closely related species. One of the key features of chromatin organization is its division into functionally distinct states—broadly classified as euchromatin, which is generally open and transcriptionally active, and heterochromatin, which is compacted and repressive. Transitions between these states, whether developmentally programmed or evolutionarily emergent, are often tightly coupled to shifts in gene regulatory activity and cellular identity.

In this study, we focus on characterizing these chromatin state transitions, with a particular interest in understanding how they are distributed across the genome and whether they exhibit predictable patterns linked to specific sequence features or regulatory elements. We ask whether certain genomic regions are more prone to undergoing structural reorganization and whether such regions harbor conserved sequence motifs, epigenetic signatures, or binding sites for key regulatory proteins. Furthermore, we investigate whether particular chromatin configurations

are disproportionately associated with genes involved in critical biological processes or with regions of the genome that show signatures of evolutionary constraint or adaptation.

By examining the diversity of chromatin states and transitions across individuals and lineages, we also aim to uncover broader principles governing chromatin architecture and its functional consequences. These insights may inform our understanding of how chromatin organization evolves, how it contributes to phenotypic diversity, and how its dysregulation may underlie disease states. Ultimately, this work contributes to a growing body of knowledge on the interplay between genome structure, function, and evolution.

1.3 A/B configurations

One widely studied aspect of chromatin architecture is its partitioning into A and B compartments, as revealed by Hi-C and other genome conformation capture techniques. These compartments represent large-scale topological domains of the genome that differ markedly in their regulatory environment and functional output. A compartments are typically associated with euchromatic regions—open, accessible chromatin that is rich in actively transcribed genes, regulatory elements like enhancers and promoters, and active histone marks such as H3K4me3 and H3K27ac. In contrast, B compartments correspond to heterochromatic, transcriptionally repressive regions characterized by reduced accessibility, fewer active genes, and enrichment of repressive epigenetic marks like H3K9me3 and DNA methylation.

This spatial segregation of the genome into A and B compartments plays a critical role in regulating which genes are expressed in a given cell or context. Genes located in A compartments are more likely to be in close physical proximity to other regulatory elements, transcriptional machinery, and nuclear compartments that promote gene activation, such as nuclear speckles. Conversely, genes embedded in B compartments are more isolated from such transcriptional hubs and may be sequestered near the nuclear lamina, where gene silencing is reinforced.

Importantly, the assignment of a genomic region to either an A or B compartment is not fixed; it can shift in response to developmental signals, environmental stress, or disease states. Such compartment switching is a powerful mechanism of gene regulation. For example, when a gene or regulatory locus moves from a B to an A compartment, it may become transcriptionally activated due to its new, more permissive chromatin environment. Likewise, genes that transition into B compartments often show downregulation or complete silencing. These dynamic shifts allow the cell to orchestrate complex gene expression programs with spatial precision, integrating structural and regulatory layers of genome function.

Moreover, comparative studies across species and populations suggest that A/B compartmentalization is both conserved and evolutionarily flexible. Certain compartments—particularly those associated with developmental and housekeeping genes—are stably maintained across evolutionary time, reflecting strong selective pressure to preserve core regulatory networks. Other regions, especially those involved in species-specific traits or environmental responses, exhibit greater compartmental plasticity. By studying the distribution and dynamics of A/B compartments,

especially in non-model organisms and hybrid genomes, we can gain novel insights into how chromatin structure evolves and how it contributes to phenotypic diversity and adaptation.

1.4 GPN

Recent advances in artificial intelligence have brought powerful tools to genomics, particularly through the development of Genomic Pre-trained Networks (GPNs)—deep learning models trained on large-scale genomic sequence data using architectures inspired by natural language processing. GPNs apply self-supervised learning approaches, analogous to models like BERT or GPT, to the DNA sequence, learning to predict masked or missing bases in context. Through this training, GPNs develop an internal representation of the “language of the genome,” capturing patterns of sequence composition, motif syntax, and long-range dependencies that underlie regulatory function.

A key advantage of GPNs lies in their use of transformer architectures, which differ significantly from traditional models like convolutional neural networks (CNNs). While CNNs are adept at recognizing short, fixed-length patterns—such as transcription factor binding motifs—they struggle with learning dependencies over long genomic distances, such as enhancer-promoter interactions or 3D chromatin looping. Transformers, in contrast, leverage self-attention mechanisms, which allow the model to compare and weigh relationships between all nucleotide positions in a sequence, regardless of distance. This makes them exceptionally well-suited for modeling the complex, multi-scale structure of regulatory DNA.

The utility of GPNs in functional genomics has been demonstrated in tasks such as predicting the effects of noncoding variants, chromatin accessibility, and transcription factor binding. For instance, the model introduced by Benegas and Batra (“DNA language models are powerful predictors of genome-wide variant effects”) showed that transformer-based GPNs outperform prior models in predicting variant impact across a variety of cell types and epigenomic contexts, even without explicit annotations. These models can generate base-resolution functional predictions across the genome, enabling insight into the likely regulatory role of any given sequence segment.

In the context of chromatin structure, GPNs hold particular promise for identifying sequence features that influence or correlate with A/B compartment identity. By learning patterns that distinguish sequences in open (A) versus closed (B) chromatin, GPNs can be used to predict how shifts in compartmentalization may arise from sequence variation. Furthermore, these predictions can be extended across species or populations to assess functional conservation—that is, whether a sequence maintains regulatory potential or compartment association despite genetic divergence.

In this study, we leverage GPN-derived embeddings and predictions to investigate the sequence determinants of chromatin organization. We aim to identify genomic regions where the GPN model predicts strong, conserved regulatory activity, and determine whether these regions also show structural stability in chromatin compartments across individuals or hybrid genomes. This approach allows us to assess whether chromatin transitions are driven primarily by changes in

DNA sequence, or whether other regulatory layers (e.g., epigenetic modifications, chromatin remodelers) are required to explain shifts between A and B compartments.

Ultimately, integrating GPN predictions with chromatin conformation data (e.g., from Hi-C) provides a powerful framework for exploring genome function at multiple scales—from base-level regulatory logic to large-scale nuclear organization. These models not only enhance our ability to interpret genomic variation and conservation but also open new avenues for studying the evolution of regulatory architecture in systems such as primates, where hybridization and population structure offer unique opportunities to dissect genome regulation in action.

1.5 **cnn vs transformer**

Convolutional Neural Networks (CNNs) have been the foundational architecture in computer vision for over a decade. Introduced by LeCun et al. (1998) and popularized with the success of AlexNet (Krizhevsky et al., 2012), CNNs exploit spatial hierarchies in image data through local receptive fields and shared weights. Architectures such as VGGNet (Simonyan & Zisserman, 2015), ResNet (He et al., 2016), and EfficientNet (Tan & Le, 2019) have progressively improved the depth, scalability, and performance of CNNs, enabling state-of-the-art results in tasks such as image classification, object detection, and semantic segmentation.

A simple Convolutional Neural Network (CNN) architecture typically consists of several key types of layers. The first are convolutional layers, which apply learnable filters (also known as kernels) to local regions of the input data. These filters are designed to capture spatial features such as edges, textures, and, as the network deepens, more complex patterns. Following the convolutional operations, activation functions are applied—most commonly the Rectified Linear Unit (ReLU)—to introduce non-linearity into the model, enabling it to learn more complex representations.

Pooling layers are another essential component of CNNs. These layers downsample the feature maps, typically using max pooling or average pooling, which reduces the spatial dimensions of the data while retaining the most salient features. This not only lowers computational requirements but also helps mitigate overfitting by providing a form of translation invariance. Toward the end of the architecture, fully connected layers are often used to perform high-level reasoning. These layers integrate the features extracted by earlier layers and are typically responsible for producing the final classification or regression outputs.

One of the main strengths of CNNs lies in their use of local receptive fields and weight sharing, which makes them highly effective at learning spatial hierarchies and translation-invariant features. Even relatively shallow CNNs can perform well on image-related tasks, particularly when training data is limited.

CNNs offer several advantages. They are parameter-efficient due to filter sharing, which significantly reduces the number of learnable weights compared to fully connected networks. Their strong inductive bias—emphasizing locality and translation invariance—makes them

particularly well-suited to visual data. Additionally, CNNs benefit from computational efficiency, as they are compatible with highly optimized GPU-based implementations, and they often require less data than more recent architectures to achieve good performance.

However, CNNs also come with limitations. They tend to focus on local features, which makes capturing long-range dependencies challenging without additional architectural mechanisms. Their receptive field is fixed and limited, so incorporating larger contextual information often requires making the network deeper or using special techniques like dilated convolutions or skip connections. Furthermore, CNN performance can be sensitive to architectural choices such as filter sizes, layer depth, and stride configurations, which often require manual tuning.

These challenges have inspired researchers to explore alternative architectures, most notably Transformer models. Unlike CNNs, Transformers are designed to model global relationships more naturally, though often at the expense of requiring more data and computational resources.

1.6 meiosis

Meiosis is the defining process of spermatogenesis, transforming diploid precursors into haploid gametes while introducing genetic diversity. The stages represented in this dataset highlight distinct steps in this transformation (Berg et al. 2019).

Spermatogonia are the mitotically active germline stem and progenitor cells at the basal compartment of the seminiferous tubules. Although they themselves do not undergo meiosis, they form the essential starting population that differentiates into spermatocytes, committing cells to the meiotic program (Berg et al. 2019).

Pachytene spermatocytes represent a central point in meiosis, specifically prophase I. During this prolonged phase, homologous chromosomes are fully synapsed through the synaptonemal complex, and crossing-over ensures genetic recombination. This is a transcriptionally active stage, producing RNAs that will later support spermatid development when transcription decreases (Berg et al. 2019).

Round spermatids are the direct products of the two meiotic divisions. They are haploid cells that no longer divide but instead undergo spermiogenesis, a process of dramatic morphological and chromatin remodeling. Although post-meiotic, they carry the molecular signatures of meiotic recombination and genome reduction (Berg et al. 2019).

During spermatogenesis, large-scale reprogramming of three-dimensional (3D) chromatin organization is required to enable meiosis and germ cell differentiation. Recent studies using low-input Hi-C in both rhesus monkeys and mice have revealed that the higher-order chromatin landscape is far from static. Instead, it undergoes stage-specific remodeling in which topologically associating domains (TADs), normally stable units of genome folding, become progressively dissolved as germ cells enter meiotic prophase. This disruption reaches its peak in pachytene spermatocytes, where TADs are almost entirely absent despite the presence of widespread transcriptional activity (Wang 2019?).

Interestingly, the loss of TADs at this stage is accompanied by the emergence of refined A/B compartments. These compartments alternate between transcriptionally active and inactive regions, providing a new level of chromatin organization that correlates more directly with gene expression than classical TAD structures. Such reorganization suggests that the meiotic genome adopts a unique 3D folding principle optimized for homologous recombination and transcriptional control (wang2019?).

A key driver of this transition appears to be the synaptonemal complex (SC), a protein scaffold that aligns homologous chromosomes during meiosis. Wang and colleagues demonstrated that in spermatocytes lacking proper SC formation, refined compartments fail to develop, and TAD-like structures re-emerge prematurely. This finding indicates that the SC does not merely stabilize chromosome pairing, but also actively contributes to global chromatin architecture by suppressing TADs and promoting compartmentalization (wang2019?).

Overall, this work highlights that spermatogenesis involves not only dynamic transcriptional and epigenetic changes but also profound reprogramming of chromatin folding. The discovery that such architectural shifts are conserved between primates and rodents underscores their biological importance. Furthermore, it raises the possibility that defects in these processes could contribute to infertility or other germline disorders, linking 3D genome organization to reproductive health (wang2019?).

1.7 X Chromosome Compartments and GC Content

One of the defining molecular features of A/B chromatin compartments is their GC content, which reflects and reinforces their functional states. A compartments are generally GC-rich and correspond to gene-dense, euchromatic regions. These areas are transcriptionally active, marked by high RNA polymerase II occupancy and enriched in active histone modifications such as H3K4me3 and H3K27ac. Their open chromatin conformation also renders them accessible to DNase I and other chromatin accessibility assays [Lieberman-Aiden et al. (2009)](Bonev and Cavalli 2016).

In contrast, B compartments tend to be AT-rich, gene-poor, and heterochromatic. These regions are associated with repressive histone marks like H3K9me3, delayed replication timing, and frequent localization near the nuclear periphery, particularly within lamina-associated domains (LADs). The compositional contrast between A and B compartments is not only a reflection of gene content but also contributes to chromatin folding, nucleosome positioning, and transcription factor binding patterns(Lieberman-Aiden et al. 2009).

Importantly, GC content remains a stable genomic feature, even as compartment identity can shift in response to developmental or environmental changes. For example, during cell differentiation or meiotic prophase, such as the pachytene stage, there is widespread chromatin reorganization, and architectural domains like TADs may temporarily dissolve. Yet, GC-rich regions consistently align with active compartments, suggesting that GC content serves as an

intrinsic guide for nuclear organization, independent of dynamic epigenetic changes (Wang et al. 2019).

This stability makes GC content a robust predictor of A/B compartment identity across cell types and developmental stages, underscoring its fundamental role in genome architecture (Wang et al. 2019).

1.8 CPG islands

CpG islands (CGIs) are typically unmethylated DNA regions that are closely associated with gene promoters and play a crucial role in regulating chromatin accessibility and three-dimensional genome organization. Accumulating evidence demonstrates that changes in CGI methylation, histone modifications, or chromatin-binding proteins can drive or accompany transitions between chromatin compartments, shifting from active A compartments to repressive B compartments or vice versa. Such transitions are frequently observed in developmental processes, cancer progression, X-chromosome inactivation, and transcriptional regulation. In many cases, CGIs act as anchors or boundaries that help define chromatin domains, and their epigenetic state can strongly influence large-scale compartmental reorganization.

Recent research on meiotic chromatin architecture provides insights that support this broader view of CGI function. A study by Wang *et al.* (2019) (Wang et al. 2019) examined spermatogenesis in rhesus monkeys and mice using low-input Hi-C and found that canonical topologically associating domains (TADs) dissolve during the pachytene stage of meiosis, even while transcription continues. This observation suggests that TAD integrity is not strictly required for gene expression at this stage. Instead of the standard large-scale A/B compartmentalization, the study identified the emergence of smaller, refined A/B compartments that alternate between transcriptionally active and inactive domains. The formation of these fine-scale compartments was shown to depend on the synaptonemal complex (SC). In SC-defective mutants, canonical TADs reappeared, indicating that the SC suppresses TADs while enabling this unique compartmental structure. Notably, this reorganization of 3D chromatin architecture was conserved between monkeys and mice, suggesting a mammalian-wide regulatory principle (Wang *et al.*, 2019) (Wang et al. 2019).

Although CpG islands were not the central focus of this study, the findings are consistent with the hypothesis that CGIs contribute to compartmental organization by functioning as boundary or anchoring elements. Because refined compartment transitions align with transcriptional activity, and CGIs are commonly associated with active transcription start sites, it is likely that they play a role in stabilizing the boundaries between alternating compartments. In this way, CGIs may provide local chromatin insulation while also influencing higher-order genome organization. (skriv om)

1.9 repeat

1.9.1 simple nested repeats

Repetitive elements make up a large portion of the eukaryotic genome and are broadly categorized into simple (tandem) and nested (interspersed) repeats. Simple repeats, such as microsatellites, minisatellites, and satellite DNA, consist of short motifs repeated in tandem and are frequently found in centromeric, subtelomeric, and regulatory regions. They are prone to replication slippage and are widely used in genetic mapping and disease diagnostics due to their high mutation rates (Gemayel et al. 2010) . In contrast, nested repeats, derived largely from transposable elements (TEs) such as LINEs, SINEs, and LTR retrotransposons, are scattered throughout the genome. These elements can insert into one another, creating complex repeat-rich regions that influence genome structure, epigenetic regulation, and chromatin organization (Bourque et al. 2018).

2 Materials and Methods

2.1 GPN architecture

The idea for this project was to evaluate the Genome-Pretrained Network (GPN) introduced in [1] and determine whether it could achieve greater accuracy than traditional methods in predicting genome-wide variant effects.

The model is designed as a convolutional neural network (CNN) and takes input sequences with a window size of 512. During training, 15% of the positions within each window are masked to enable the model to learn meaningful representations.

The architecture consists of 25 layers, each structured as follows: a dilated convolution layer, followed by an add-and-norm layer with a skip connection from before the dilated convolution. This is followed by a feedforward layer, another add-and-norm layer, and additional skip connections.

A feedforward layer is a fundamental component of neural networks, where inputs pass through one or more fully connected layers with activation functions, transforming data without looping back. This structure helps the model learn complex representations by applying weighted transformations and non-linearities.

A dilated convolution expands the receptive field of a convolutional layer without increasing the number of parameters or reducing resolution. By spacing kernel elements apart, it captures long-range dependencies in sequences, making it particularly useful in genomic data analysis. When combined, dilated convolutions and feedforward layers enhance a model's ability to recognize patterns across different scales efficiently.

After passing through the 25 layers, the model produces a contextual embedding with a dimension of 512 ($D=512$), followed by classification layers. The final layer outputs the probabilities of the four nucleotides at each masked position.

The GPN variant effect prediction score is calculated as the log-likelihood ratio between the alternate (ALT) and reference (REF) alleles. Here, L represents the window length in base pairs, and D denotes the embedding dimension.

2.2 Data Preparation and Pretraining Workflow

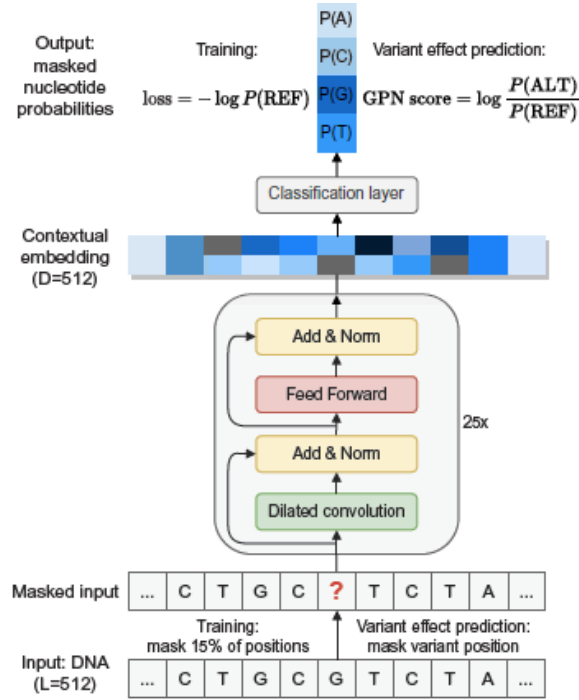


Figure 2.1

2.2 Data Preparation and Pretraining Workflow

To evaluate the Genome-Pretrained Network (GPN), the initial step involved reproducing the pretraining workflow provided in the original publication. This workflow outlines the procedure for generating a dataset compatible with the model, specifically for masked language modeling over genomic sequences.

Initial attempts to apply the workflow to a baboon genome dataset encountered technical difficulties. These issues were not intrinsic to the baboon data itself but stemmed from the implementation of the workflow. To ensure compatibility with the model and maintain consistency with the original study, the Anatoptis genome—used as the reference in the published work—was selected as a pilot dataset.

After resolving the implementation issues, a dataset suitable for GPN pretraining was successfully generated. The resulting data consisted of masked genomic sequences, formatted according to the model's input requirements.

2.3 Model Execution Workflow and Hardware Constraints

Following the successful generation of a pretraining-compatible dataset, the next step involved attempting to run the Genome-Pretrained Network (GPN) using the available inference and training workflows. The documentation provided on the model’s GitHub repository lacked detailed guidance, making it challenging to understand the complete execution process and the rationale behind certain steps.

To proceed, an alternative workflow was identified—one specifically designed for use with the Anatoptis genome. This workflow was adopted as a reference implementation for executing the GPN model.

However, significant hardware limitations emerged during this phase. The original study reported using four NVIDIA A100 GPUs, each equipped with 80 GB of RAM. In contrast, the available system utilized NVIDIA L43R GPUs with 45 GB of RAM, resulting in substantial differences in computational capacity.

As a result, attempts to execute the GPN model encountered frequent out-of-memory (OOM) errors, particularly during forward passes through the deeper layers of the network. These memory constraints posed a major obstacle and significantly hindered reproducibility under the available hardware setup.

Initial efforts to train or fine-tune the Genome-Pretrained Network (GPN) focused on applying the workflow to a baboon genome using a multi-GPU setup. The goal was to replicate the original training environment, which employed four NVIDIA A100 GPUs (80 GB RAM each). However, the available infrastructure utilized NVIDIA L43R GPUs, each with 45 GB of RAM, which introduced several compatibility and performance challenges.

Early testing began with the Anatoptis genome. Attempts to process the entire genome using the Snakemake workflow were unsuccessful due to GPU detection issues. These issues stemmed from the cluster configuration, which prevented the Snakemake environment from correctly interfacing with the available GPU hardware.

After resolving these initial configuration issues, the workflow was executed with GPU support. While this improved performance, the model began encountering memory errors after a few hours of training, even when utilizing two to four GPUs in parallel. In an effort to overcome these constraints, the number of GPUs was incrementally increased—from four to six, and finally to eight—providing an aggregate of 360 GB of GPU RAM.

Despite these efforts, the model continued to encounter out-of-memory (OOM) errors during execution. Even with eight GPUs, the memory demands of the GPN model remained prohibitive under the given cluster configuration. These persistent failures highlighted a significant limitation: the model’s architecture and resource demands exceeded the practical limits of the available hardware.

Ultimately, these constraints prevented successful execution of the GPN model on the tested system.

Given the persistent memory limitations encountered during full-genome processing, an alternative strategy was implemented. Rather than inputting the entire Anatoptis genome, the data was partitioned at the chromosome level. This allowed individual chromosomes to be processed independently, while still allocating the full available GPU resources (eight NVIDIA GPUs with a combined 360 GB of RAM) to each run.

This approach proved successful. When processing a single chromosome at a time, the GPN model was able to execute without encountering memory-related failures. As a result, contextual embeddings were successfully generated for individual chromosomes of the Anatoptis genome.

However, due to time constraints near the conclusion of the project timeline, the extracted embeddings were not subjected to further downstream analysis. Although they were successfully retrieved and stored, no additional modeling, interpretation, or evaluation steps were completed within the project scope.

2.4 using HIC to extract transtion areas

2.5 DATA

The dataset used in this study was generated from *Macaca* (rhesus macaque) samples, covering five distinct stages of spermatogenesis, as well as fibroblasts for comparison. These stages represent the continuum of male germ cell development from undifferentiated precursors to mature gametes.

Spermatogonia are the mitotically active stem and progenitor cells located at the basal layer of the seminiferous tubules. They serve as the foundation of spermatogenesis by both self-renewing to maintain the stem cell pool and differentiating to give rise to spermatocytes. This stage is characterized by robust proliferative activity and the initiation of gene expression programs that prepare cells for meiosis.

Pachytene spermatocytes are primary spermatocytes in the prolonged pachytene stage of meiotic prophase I. During this phase, homologous chromosomes are fully synapsed and undergo genetic recombination through crossing-over, ensuring genetic diversity in gametes. These cells also exhibit high levels of transcriptional activity, producing RNAs that will be used later during spermatid development, as transcription becomes reduced after meiosis.

Following meiosis II, round spermatids are formed as haploid germ cells. At this point, the cells no longer divide but instead undergo extensive remodeling during spermiogenesis. Chromatin begins to compact, transcriptional activity decreases, and transcripts stored earlier are translated to support structural changes. Morphologically, the cells remain spherical at this stage.

Through spermiogenesis, round spermatids differentiate into elongated spermatids and ultimately mature spermatozoa. This process involves dramatic nuclear condensation, replacement of

2 Materials and Methods

histones with protamines, formation of the acrosome, and development of the flagellum for motility. Mature sperm are transcriptionally silent and highly specialized for fertilization.

Finally, fibroblasts, isolated from connective tissue, were included as a non-germline somatic control. Unlike germ cells, fibroblasts are not involved in reproduction but maintain structural and extracellular matrix functions. Their inclusion allows for distinguishing germline-specific molecular signatures from general cellular processes.

Together, these samples provide a comprehensive view of germ cell differentiation across spermatogenesis, with fibroblasts serving as a reference point for somatic baseline comparisons.

2.6 Datahandling of the data befor analysisi

To load the data we use the function `parse_compartment_data(file_name)`, which transforms the raw eigenvector track into meaningful genomic segments. The function begins by reading a CSV file that contains at least one column, `e1`, representing eigenvector values at 100 kb resolution.

The eigenvector track (E1) marks A/B compartments along the genome. During its generation, some bins may contain NaN values. If these are left in the dataset, each missing value is treated as a break and produces false edges, creating compartment transitions that do not actually exist. To avoid this problem, we remove all rows containing NaN values before proceeding:

```
e1_100kb.dropna(inplace=True)
```

This step ensures that the sign changes in the eigenvector reflect only genuine compartment transitions.

After cleaning, the function calculates genomic coordinates by assigning each bin a start and end position based on its index and the fixed bin size of 100 kb. With these coordinates in place, it determines the sign of each eigenvector value and assigns segment IDs whenever the sign changes, so that continuous stretches of positive or negative values are grouped into compartments.

Each compartment is then summarized by calculating the mean and sum of its eigenvector values and recording the minimum start and maximum end positions. Segments with a positive mean are classified as compartment A, while those with a negative mean are classified as compartment B. To prevent artifacts caused by missing bins, the function also checks segment borders to ensure that gaps do not create artificial extents.

The function produces two outputs. The first is a tidy data frame of compartments that includes chromosome, start and end positions, mean eigenvector value, segment ID, sign, and compartment label. The second is a sorted list of all unique segment boundaries, which represent the genomic positions of compartment transitions.

To separate the data into A and B compartments, we applied a simple filtering step. For each genomic interval, we compared the compartment label (A or B) with the sign of its position relative to the nearest transition. Intervals that matched the expected direction were collected into one set, while those that fell on the opposite side were collected into another. In practice, this was done with the following code:

```
A_val = result[
    ((result['comp'] == 'A') & (result['start'] < 0)) |
    ((result['comp'] == 'B') & (result['start'] > 0))
].copy()
B_val = result[
    ((result['comp'] == 'A') & (result['start'] > 0)) |
    ((result['comp'] == 'B') & (result['start'] < 0))
].copy()
```

This logic ensures that the data is consistently separated into values supporting the A compartment and values supporting the B compartment, based on both the compartment label and the position relative to the transition point.

2.7 Distance between the edges

To evaluate the availability of data across genomic distances and mitigate potential bias in downstream analyses, we calculated reverse cumulative counts of compartment edge half-distances. First, edges were identified at compartment transitions between A and B (in either direction). For each consecutive pair of such edges, we measured the genomic distance between their start positions and divided it by two, yielding the half-distance between edges.

We then binned these half-distances and computed the reverse cumulative count for each bin, which represents the number of edges with half-distances greater than or equal to that bin. This approach effectively measures the number of edges capable of contributing data at a given spatial scale. Using this metric allows us to normalize other measurements by the available data density, reducing the risk that apparent trends are driven by varying numbers of contributing edges at different distances rather than true biological effects. Counts were calculated up to 2 Mb to encompass the maximum observed separation.

2.8 compare GC content to transition in macaca

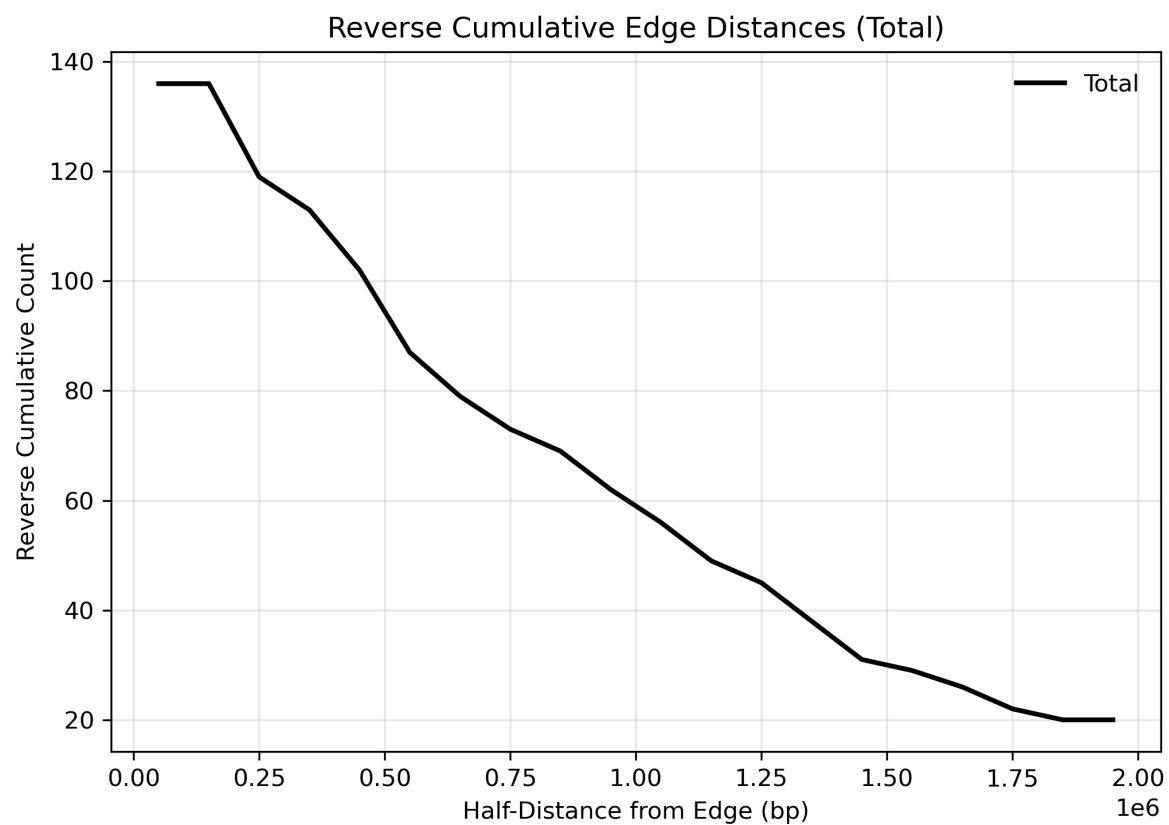


Figure 2.2: Reverse cumulative counts of compartment edge half-distances..

3 result

3.1 compare GC content to transition in macaca

We compared GC content between A and B chromatin compartments for chromosome 8 and chromosome X across five Macaca cell types: fibroblasts, sperm, round spermatids, pachytene spermatocytes, and spermatogonia. Chromatin compartments were defined from A–B and B–A structural transitions, and GC content was measured separately for each compartment.

For chromosome 8, the A compartment consistently showed higher GC content than the B compartment in all cell types, which aligns with expected genomic organization. In chromosome X, however, the differences between compartments were smaller. In round spermatids, GC content was nearly identical between A and B compartments, while in pachytene spermatocytes, spermatogonia, and sperm, A and B compartments had closely overlapping GC values. Interestingly, in fibroblasts the pattern was reversed, with the B compartment displaying higher GC content than the A compartment see Figure 6.1.

3.2 CPS islands

To investigate chromatin dynamics during spermatogenesis in Macaca, we analyzed genome-wide changes in accessibility across successive developmental stages. We focused on genomic regions undergoing transitions from open to closed chromatin states and vice versa, as these sites represent potential regulatory hotspots that may contribute to germ cell differentiation. To explore the relationship between these transitions and CpG-rich elements, we obtained a genome-wide track of CpG islands and mapped their positions relative to the transition regions. Each transition was further annotated according to its placement within A or B compartments, with compartment A typically associated with transcriptionally active euchromatin and compartment B corresponding to transcriptionally repressed heterochromatin. By integrating chromatin transitions, CpG island mapping, and compartmentalization, we generated a stage-specific view of genome organization across spermatogenesis.

Figure 3.2 illustrates the distribution of CpG island-associated transitions in fibroblasts on chromosome 8. At distances up to ~0.5 Mb, transition counts were comparable between A and B compartments, but beyond this range the A compartment exhibited substantially higher counts. This enrichment persisted until ~1.5 Mb, after which both compartments declined toward zero. The drop at larger distances likely reflects technical constraints: to measure a

3 result

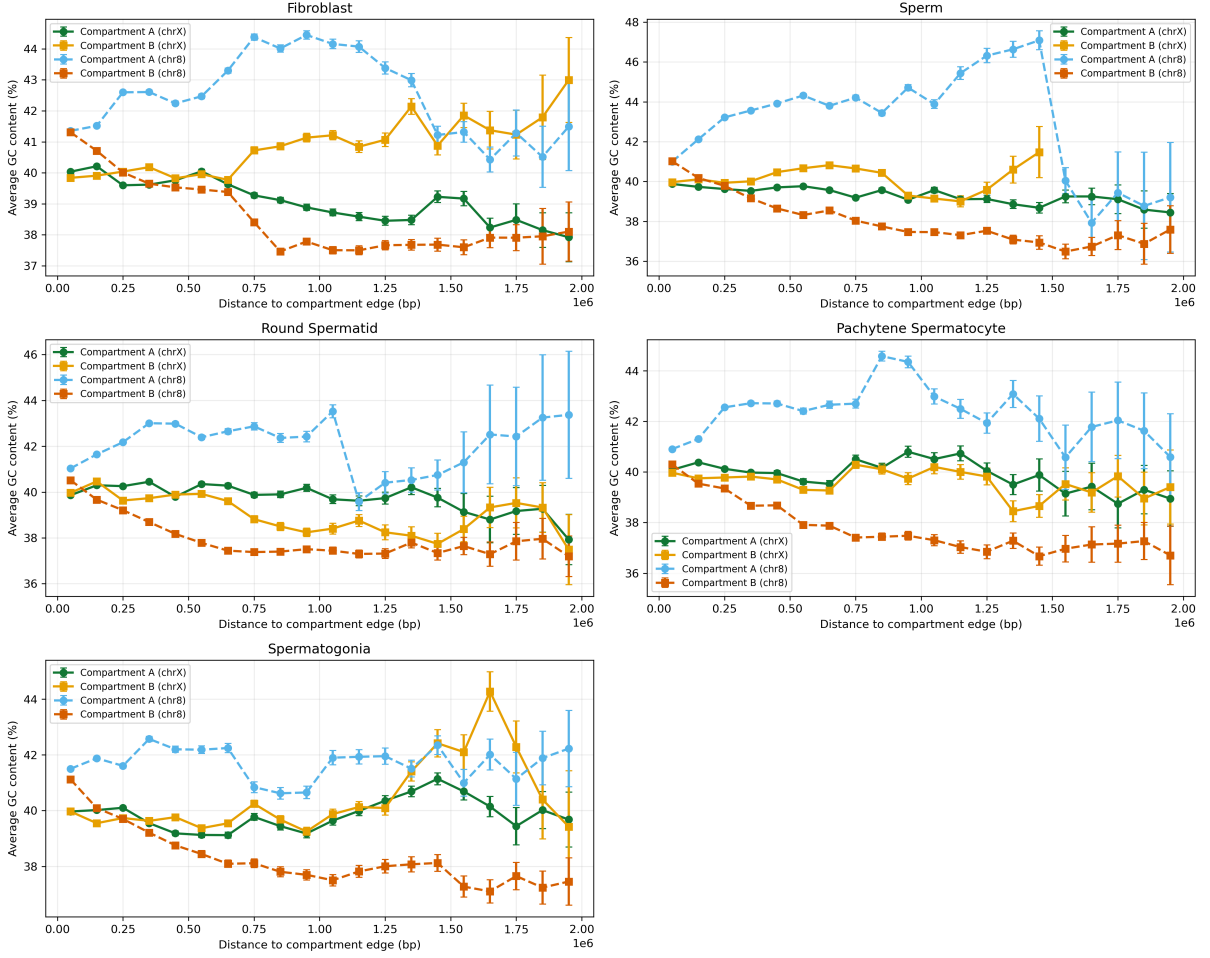


Figure 3.1: Compartment analysis for chromosomes X and 8 in five *Macaca* cell types—fibroblasts, sperm, round spermatids, pachytene spermatocytes, and spermatogonia. GC content was remapped to genomic locations where chromatin structure transitions between compartments A and B. The data were separated so that each compartment contains only A or B structure. For each genomic bin, the plot shows the mean GC content, with error bars representing the standard deviation of the mean, adjusted by the effective edge contribution (accounting for how many edges contribute at longer distances). For chromosome 8, the A compartment exhibits higher GC content than the B compartment. In tissues involved in spermatogenesis, chromosome X shows similar GC content in both A and B compartments. In fibroblasts, GC content is higher in the B compartment than in the A compartment.

transition at 1.5 Mb, the minimum spacing between sites must be 3 Mb, which is relatively rare across the genome. This limitation is evident even though counts were normalized to account for the number of possible contributing sites. By contrast, chromosome X showed no compartment-specific differences in fibroblasts, with A and B compartments displaying similar counts across all distances.

In spermatogonia, no compartment-specific differences were observed on either chromosome 8 or chromosome X. Transition counts remained comparable between A and B compartments across all distances, in contrast to the fibroblast profile where chromosome 8 showed A-compartment enrichment.

In pachytene spermatocytes, chromosome 8 displayed a distinct compartmental pattern. Between 0 and ~1.25 Mb, transition counts were higher in the A compartment compared to the B compartment, but at longer distances the two converged. On chromosome X, however, A and B compartments remained similar, showing no evidence of compartment-specific enrichment.

In round spermatids, chromosome 8 again showed a bias toward the A compartment, with more transitions observed from 0 to ~1.1 Mb. A comparable, though slightly shifted, pattern was seen on chromosome X, where enrichment in the A compartment was detected between ~0.25 and 1.3 Mb.

Finally, in mature sperm, chromosome 8 exhibited consistent A-compartment enrichment from 0 to ~1.5 Mb. On chromosome X, the effect was more modest, with only a slight increase in A-compartment counts between ~1.0 and 1.75 Mb.

Together, these results reveal stage-specific differences in the relationship between chromatin transitions, CpG islands, and genome compartmentalization. Fibroblasts and spermatogonia lack strong compartmental biases, while later spermatogenic stages, particularly pachytene spermatocytes, round spermatids, and mature sperm, display pronounced enrichment of CpG island-associated transitions in the A compartment, especially on chromosome 8.

3.3 simple repeat

We next examined the relationship between chromatin transitions and simple repeats using the same strategy as for CpG islands. The genomic positions of simple repeats were remapped relative to transition sites, and the counts in each distance bin were normalized by the number of transition sites that could contribute to that bin. This approach allowed us to determine whether simple repeats are enriched near transition regions and whether their distribution differs between A and B compartments.

In fibroblasts, chromosome 8 showed a slight enrichment of simple repeats in the B compartment compared to the A compartment. On chromosome X, however, the opposite pattern was observed, with more simple repeats detected in the A compartment. Across both chromosomes, simple repeats were generally more abundant close to transition sites, with their frequency declining as distance increased.

3 result

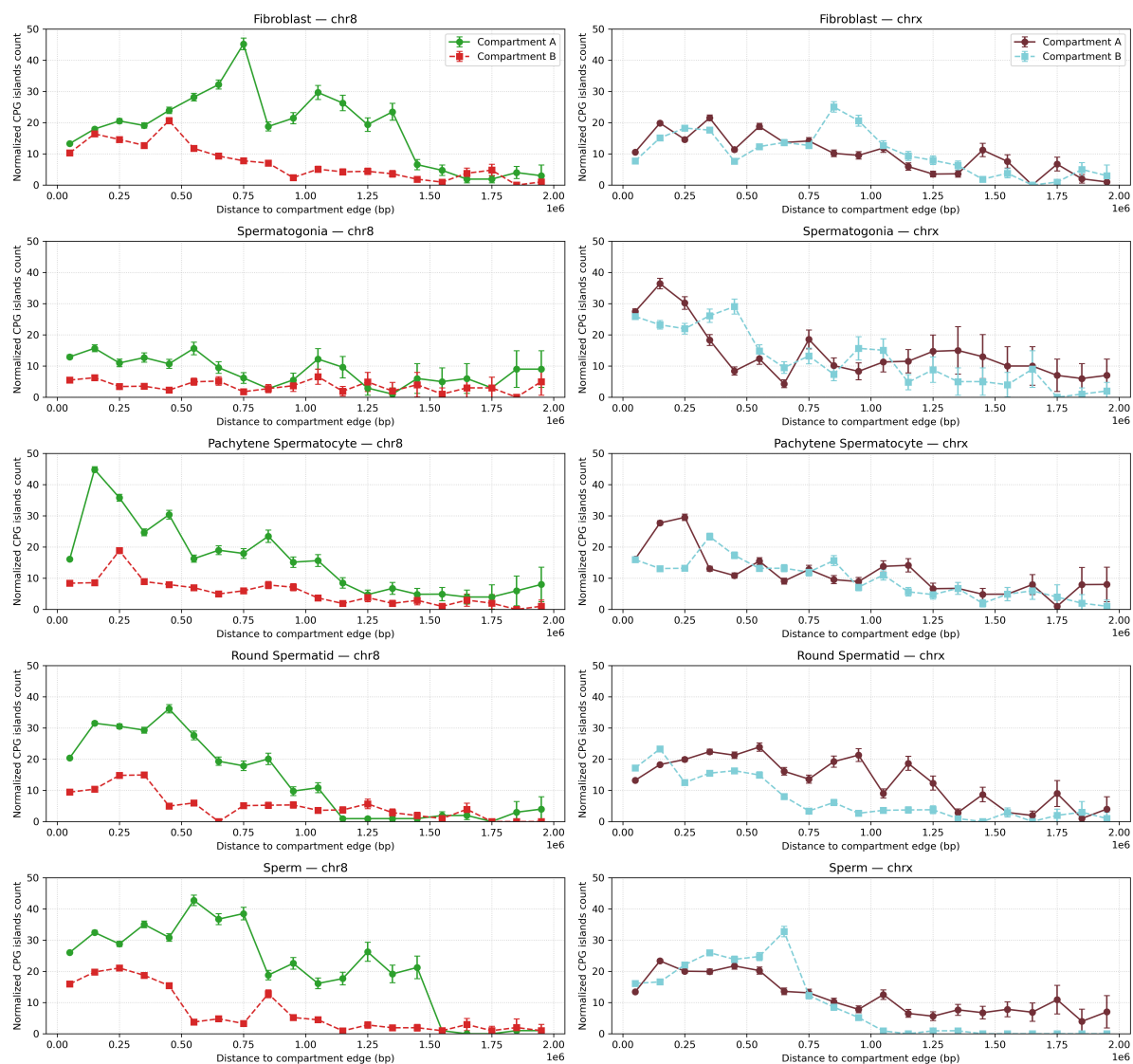


Figure 3.2: **Figure:** cpgislands

In spermatogonia, chromosome 8 did not display any clear compartment-specific differences, as simple repeat counts were comparable between A and B compartments across all distances. On chromosome X, simple repeats were evenly distributed between compartments up to ~0.5 Mb, but from ~0.5 to 2 Mb the A compartment contained higher counts. Similar to fibroblasts, spermatogonia also showed strong enrichment of simple repeats near transition sites, with counts decreasing at greater distances.

In pachytene spermatocytes, chromosome 8 again showed more simple repeats in the B compartment than in the A compartment, with a strong enrichment close to transition sites. On chromosome X, an A-compartment bias emerged beyond ~0.75 Mb, where more simple repeats were observed in A compared to B.

In round spermatids, chromosome 8 displayed a consistent bias toward the B compartment, with more simple repeats present there than in the A compartment. By contrast, chromosome X showed the opposite trend, with higher counts of simple repeats in the A compartment.

Finally, in mature sperm, chromosome 8 continued to show an enrichment of simple repeats in the B compartment, while chromosome X again displayed more simple repeats in the A compartment.

3.4 nested repeat

We also analyzed the distribution of nested repeats relative to chromatin transitions using the same mapping and normalization strategy applied to CpG islands and simple repeats. The results for nested repeats closely paralleled those observed for simple repeats.

In fibroblasts, chromosome 8 showed more nested repeats in the B compartment, while chromosome X displayed a bias toward the A compartment. In spermatogonia, chromosome 8 showed no compartment-specific differences, whereas chromosome X exhibited an A-compartment enrichment beyond ~0.5 Mb. Pachytene spermatocytes, round spermatids, and mature sperm followed the same overall trends: chromosome 8 consistently contained more repeats in the B compartment, while chromosome X showed enrichment in the A compartment. Across all stages and both chromosomes, nested repeats were most abundant near transition sites, with counts decreasing at larger distances.

3 result

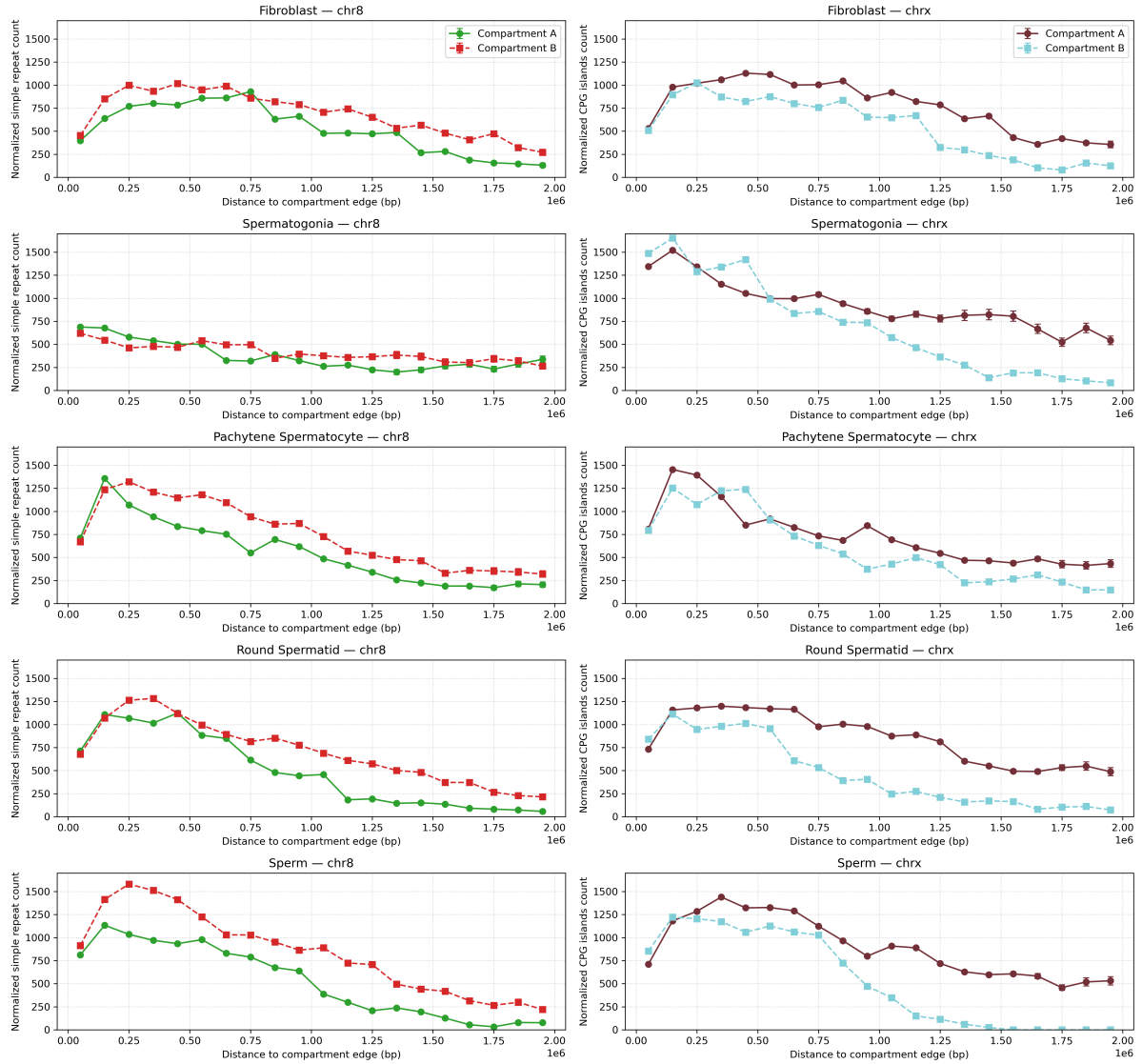


Figure 3.3: **Distribution of simple repeats relative to chromatin transition sites in fibroblasts and spermatogenic stages.**

Chromosome 8 is shown on the left and chromosome X on the right. Rows represent different cell types: fibroblasts, spermatogonia, pachytene spermatocytes, round spermatids, and mature sperm. In each panel, simple repeat counts were mapped relative to transition sites, and values in each distance bin were normalized by the number of transition sites that could contribute to that bin.

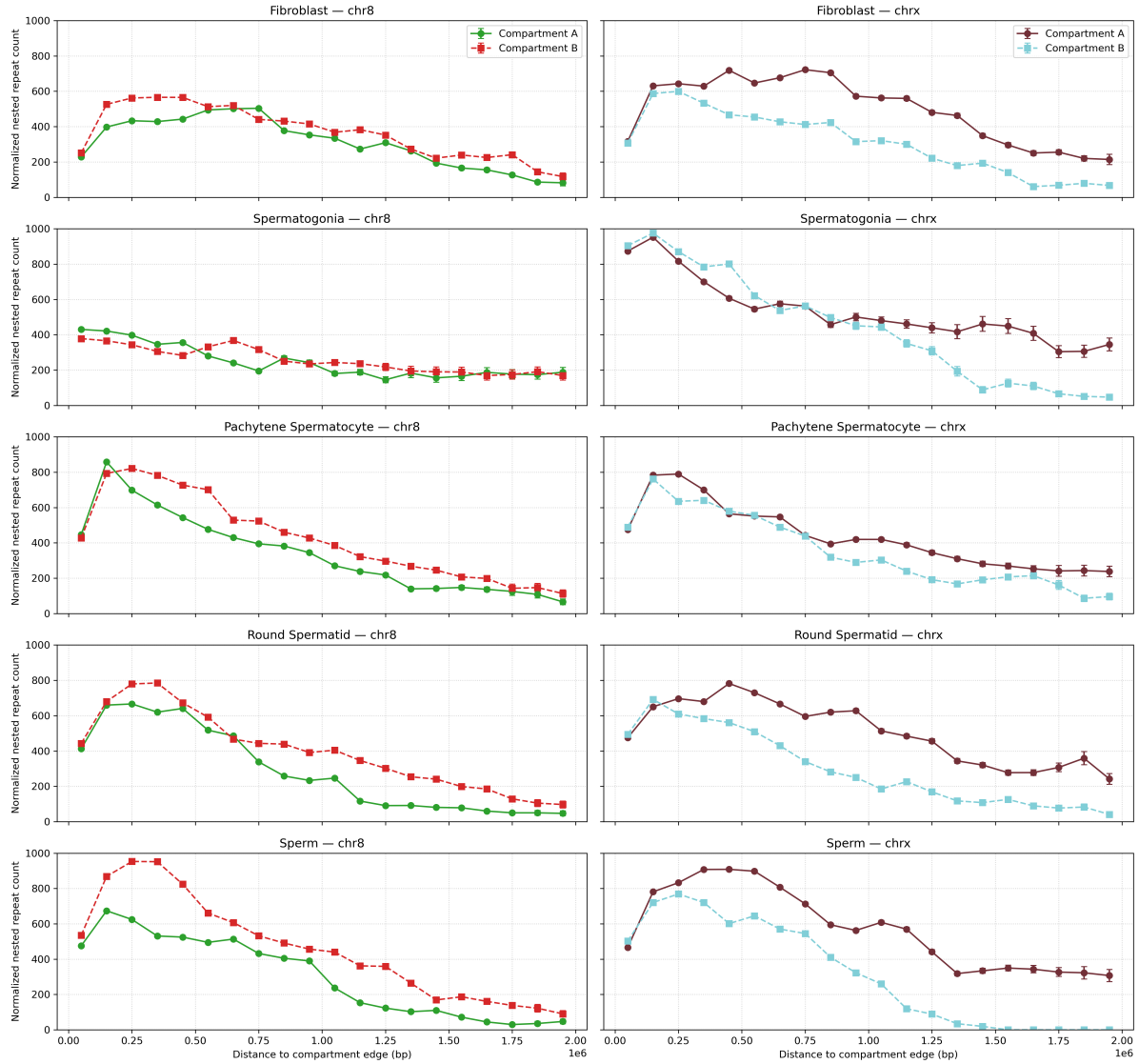


Figure 3.4: **Distribution of nested repeats relative to chromatin transition sites in fibroblasts and spermatogenic stages.**

Chromosome 8 is shown on the left and chromosome X on the right. Rows represent different cell types: fibroblasts, spermatogonia, pachytene spermatocytes, round spermatids, and mature sperm. In each panel, nested repeat counts were mapped relative to transition sites, and values in each distance bin were normalized by the number of transition sites that could contribute to that bin.

4 Dissusion

4.1 repeats

5 Conclusion

6 maybe artikler



Figure 6.1: Compartment analysis for chromosomes X and 8 in five Macaca cell types—fibroblasts, sperm, round spermatids, pachytene spermatocytes, and spermatogonia. GC content was remapped to genomic locations where chromatin structure transitions between compartments A and B. The data were separated so that each compartment contains only A or B structure. For each genomic bin, the plot shows the mean GC content, with error bars representing the standard deviation of the mean, adjusted by the effective edge contribution (accounting for how many edges contribute at longer distances). For chromosome 8, the A compartment exhibits higher GC content than the B compartment. In tissues involved in spermatogenesis, chromosome X shows similar GC content in both A and B compartments. In fibroblasts, GC content is higher in the B compartment than in the A compartment.



Figure 6.2: **Figure:** cpgislands

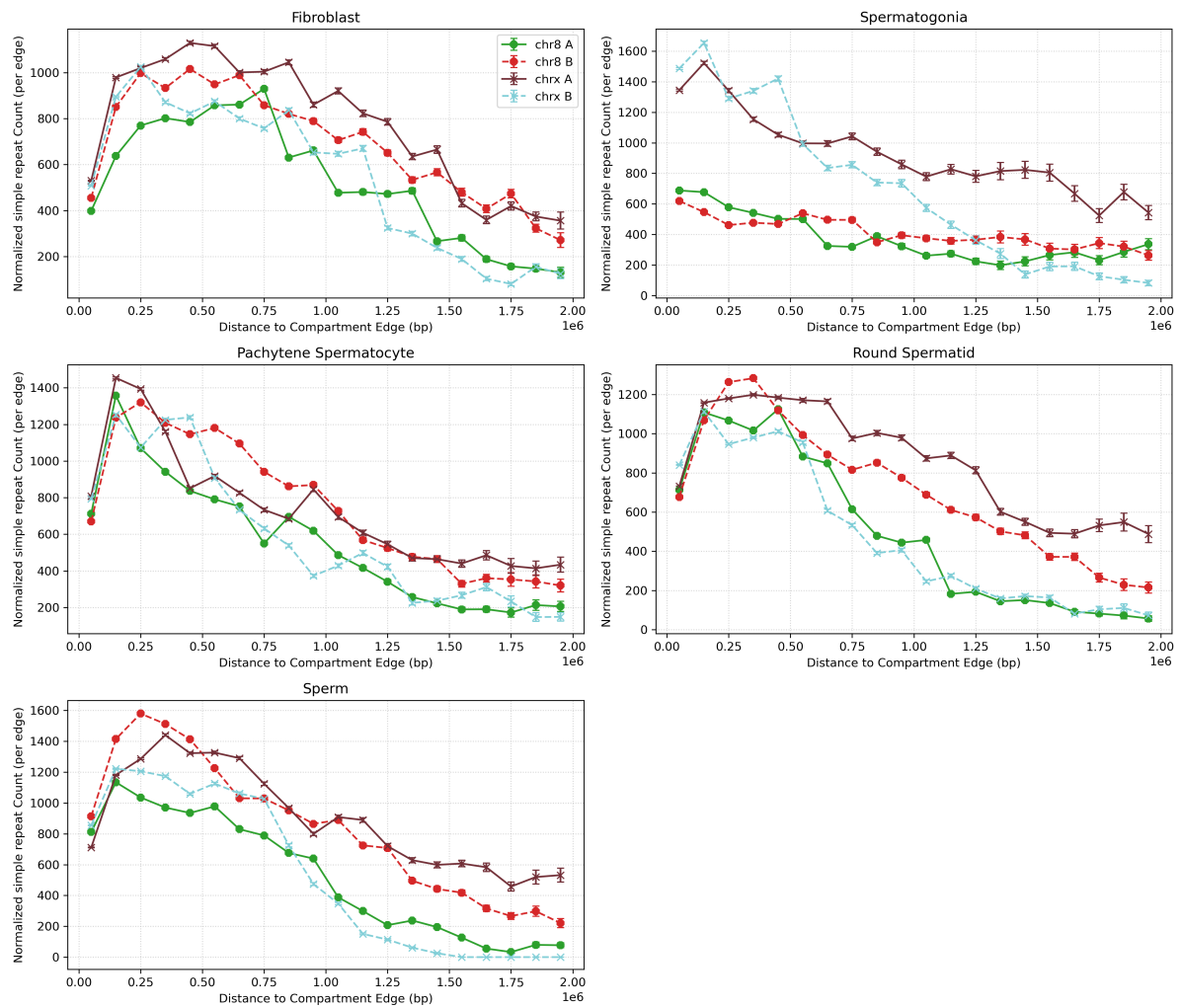


Figure 6.3: **Figure:** simple_repeat repeat

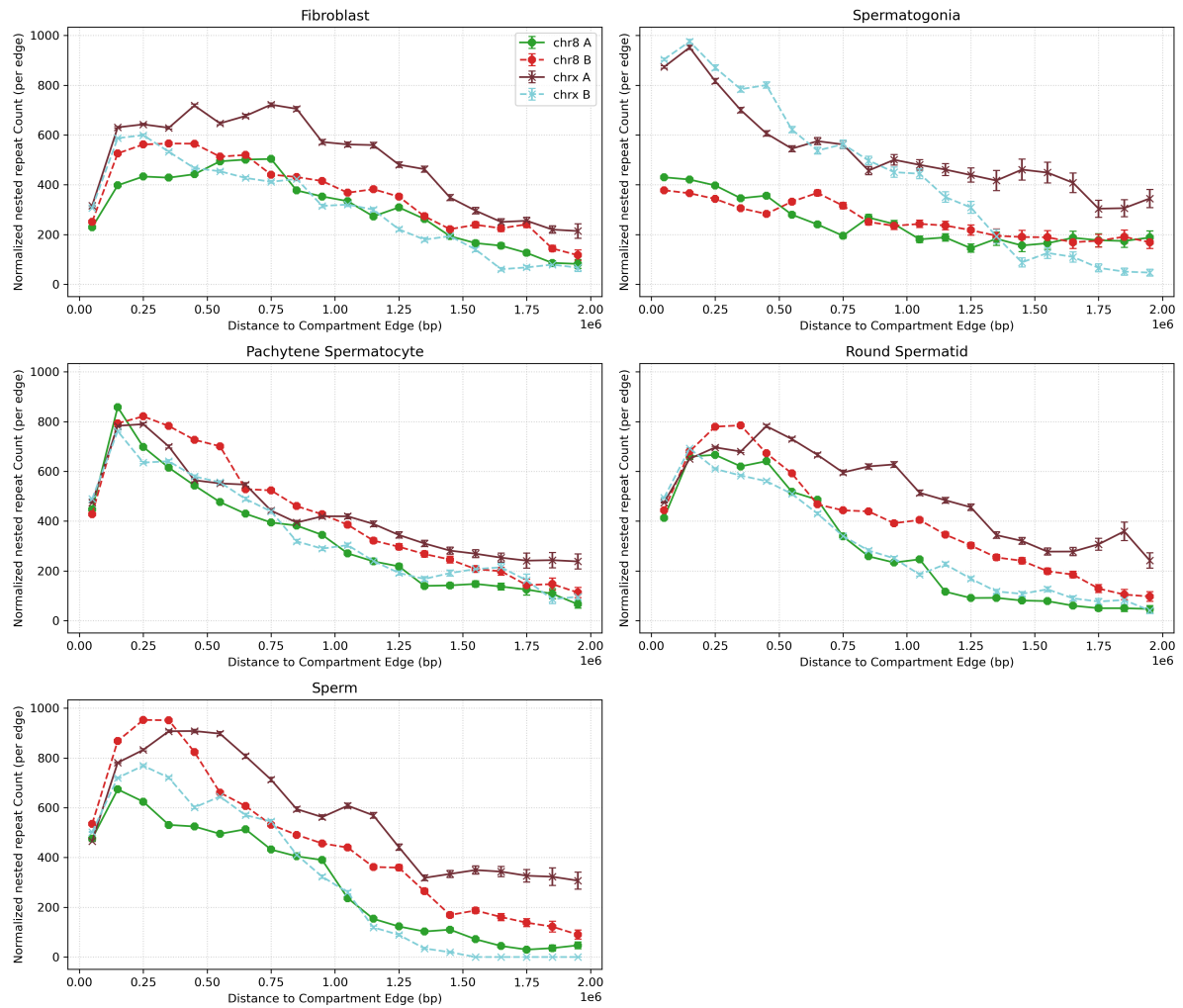


Figure 6.4: **Figure:** nested repeat

7 suppelementery

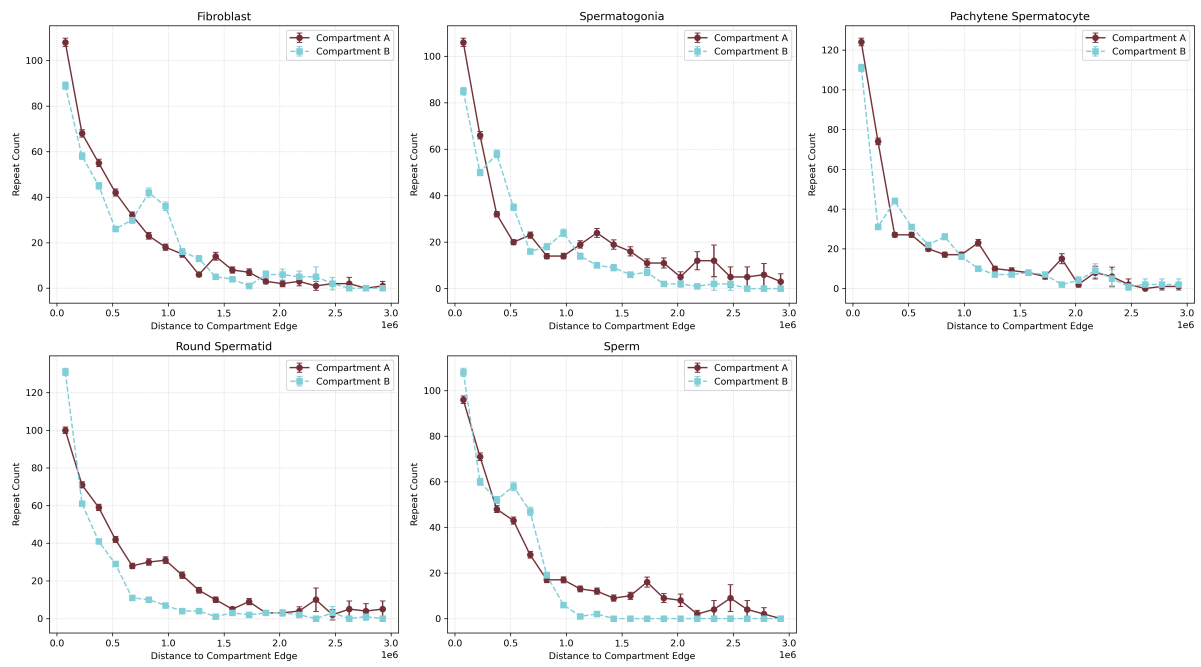


Figure 7.1: chrX

7.1 chr 8

7 supplementary

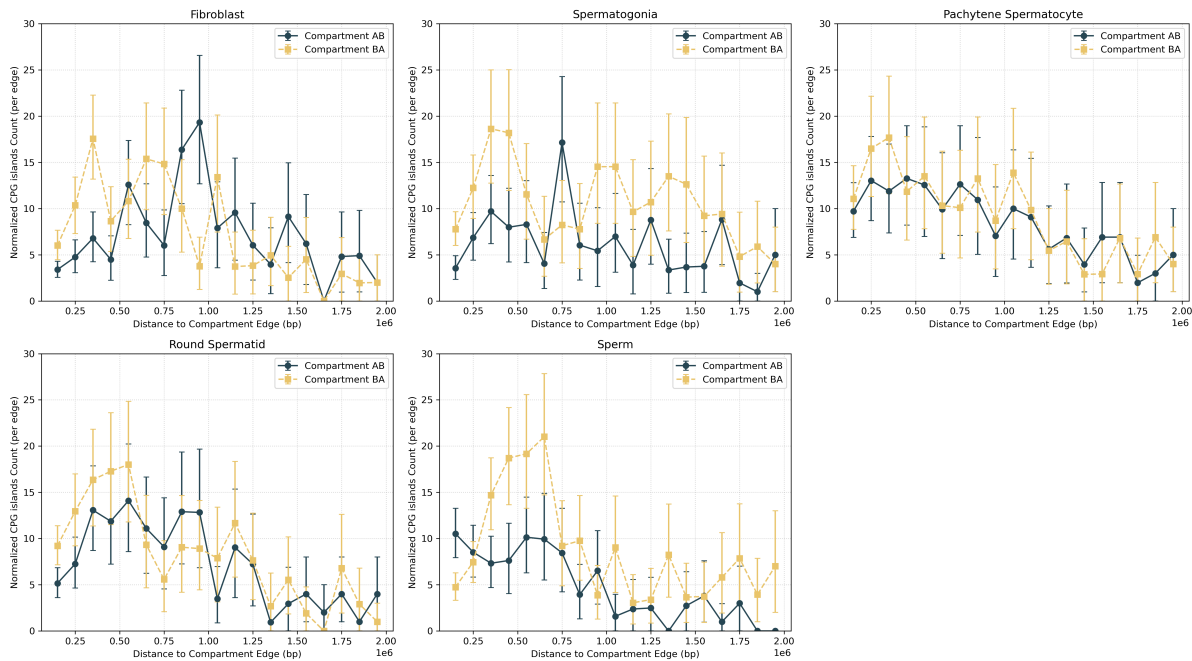


Figure 7.2: chrX

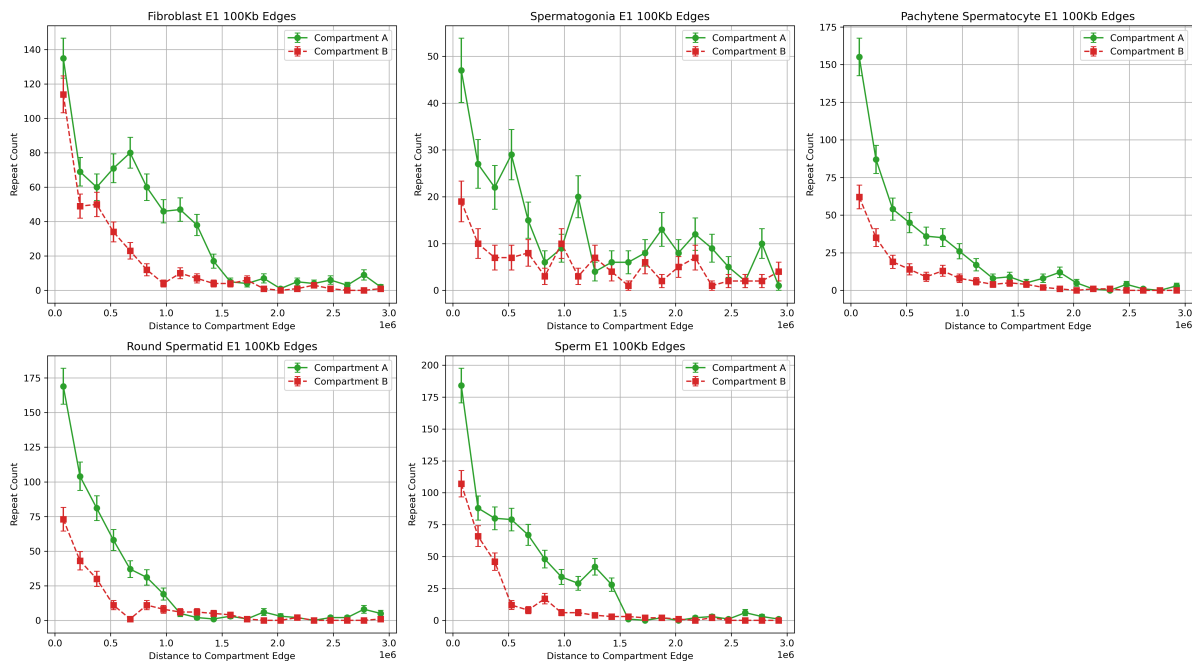


Figure 7.3: chr8

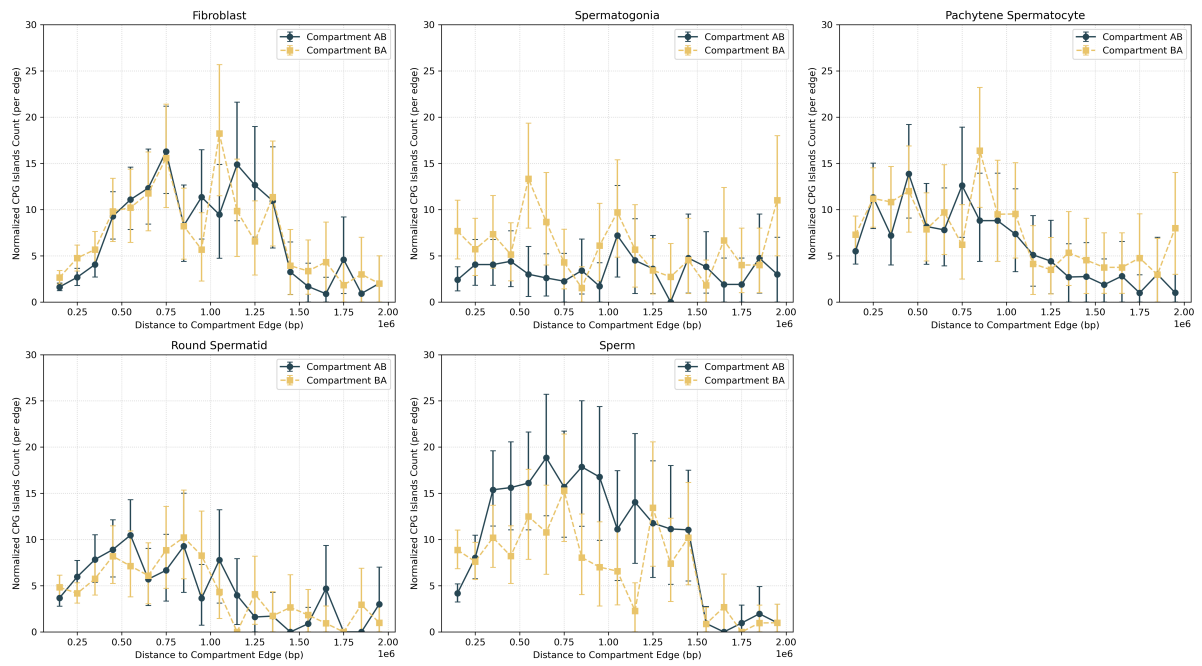


Figure 7.4: chr8

8 notes

9 chromatin structure and regulation of genes

2. Regulation of Gene Expression The chromatin state influences gene accessibility:

Open chromatin is associated with active gene transcription.

Closed chromatin represses gene activity. Repetitive elements near regulatory genes can shift chromatin structure, acting as silencers or enhancers depending on the epigenetic landscape (Venkatesh & Workman, 2015).

Venkatesh & Workman (2015) – Histone exchange, chromatin structure and transcription regulation *Nature Reviews Molecular Cell Biology* Highlights the plasticity of chromatin around repeats and its impact on transcription.

10 repeats surrounding edges

1. Genome Stability and Repression of Transposable Elements Repetitive sequences, such as LINEs, SINEs, and satellite DNA, are often silenced via heterochromatin formation. This closed chromatin state is essential to suppress transposable elements, which can otherwise mobilize and disrupt genome integrity (Moazed, 2011).

Makova & Hardison (2015) – The effects of chromatin organization on variation in mutation rates Nature Reviews Genetics Discusses how heterochromatin protects repeats but also contributes to mutational heterogeneity.

4. Development and Cell Differentiation During development, regions with repetitive DNA undergo reconfiguration. The switch from euchromatin to heterochromatin is essential for cell lineage specification, and misregulation can cause developmental disorders (Gelato & Fischle, 2008).

Ghosh & Woodcock (2010) – Chromatin higher-order structure and dynamics Cold Spring Harbor Perspectives Explains how chromatin compaction is governed by nucleosome positioning and repeat regions.

10.0.1 Three-dimensional genome reorganization during mouse spermatogenesis

- Highlights how A/B compartment transitions coincide with major cell state transitions during spermatogenesis.
- Found TF motifs and simple repeats enriched in switching regions.
- Sheds light on regulatory elements linked to chromatin phase switching.

Core Findings This study investigates how the 3D organization of the genome changes across seven sequential stages of mouse spermatogenesis, using Hi-C technology. The work specifically focuses on topologically associating domains (TADs), chromatin loops, and A/B compartments, revealing highly dynamic reorganization events.

1. TAD and Loop Dynamics Present in Type A spermatogonia (early stage).

Disappear at pachytene stage (meiosis I prophase).

Reappear in mature spermatozoa.

This pattern indicates that TADs are not static during development and may not be essential for transcription at all stages.

2. CTCF Binding CTCF remains bound at TAD boundary regions even when TADs are absent (e.g., in pachytene spermatocytes).

This suggests CTCF binding is not sufficient by itself to maintain 3D structure.

3. Enhancers, Promoters, and Active Transcription Despite the absence of TADs at pachytene, enhancers and promoters retain open chromatin.

Transcription continues, implying that gene expression can occur independently of higher-order chromatin loops.

4. A/B Compartment Changes A/B compartments are largely conserved on autosomes throughout spermatogenesis.

However, specific A → B switching events are correlated with changes in gene activity.

The X chromosome loses its A/B compartment structure during stages like pachytene spermatocytes (pacSC), round spermatids (rST), and elongating spermatids (eST).

Repeats and Chromatin Compartment Transitions TF motif enrichment was analyzed in differentially active chromatin regions, which also included repetitive DNA content.

During A/B compartment switching, particularly in regions transitioning from B (closed) to A (open), some simple and low-complexity repeats were enriched, suggesting a role in:

Regulatory rewiring during cell fate transition.

Chromatin accessibility and transcription activation.

The X chromosome, which undergoes massive reorganization and loss of A/B compartment structure during specific stages (pacSC, rST, eST), may exhibit a distinct repeat signature, though this was only briefly hinted at in the paper.

(**article?**) {Luo2019, year = {2019}, title = {{Three-dimensional genome reorganization during mouse spermatogenesis}}, author = {Luo, Zhen and Wang, Xin and Wang, Rui and Chen, Jie and Chen, Yan and Xu, Qing and Cao, Jun}, journal = {bioRxiv}, doi = {10.1101/585281}, abstract = {{Dynamic changes in chromatin architecture accompany spermatogenesis. Using Hi-C and ATAC-seq, this study reveals A/B compartment transitions and transcription factor motif enrichments that align with germ cell stage-specific chromatin changes.}}, pages = {}, number = {}, volume = {}, pmid = {}, pmcid = {}, issn = {}, keywords = {}, local-url = {https://www.biorxiv.org/content/10.1101/585281v1} }

10.0.2 Single-cell long-read Hi-C, scNanoHi-C2

- Uses single-cell Hi-C to resolve chromatin changes in early germ cells.
- Finds TE subfamilies enriched differently in A/B compartments, suggesting a regulatory function of repeat elements.

Key Findings Summary Development of scNanoHi-C2

The authors introduce scNanoHi-C2, a novel single-cell long-read Hi-C technique.

It overcomes limitations of standard Hi-C by capturing long-range, multi-contact chromatin interactions in individual cells.

3D Genome Dynamics in Germ Cells

Applied to embryonic germline cells, the technique reveals:

Stage-specific topological reorganization of the genome.

Highly dynamic A/B compartment transitions, even at early embryonic stages.

Repeat Element Profiling

The study shows distinct enrichment patterns of transposable elements (TEs) in A and B compartments.

LINEs and SINEs were differentially represented, correlating with developmental chromatin remodeling.

These repeats may contribute to insulation, compartmental identity, or nuclear architecture.

Chromatin Rewiring during Primordial Germ Cell (PGC) Development

Early embryonic germ cells exhibit weaker TAD boundaries and less stable compartments compared to somatic cells.

As differentiation progresses, compartment strength increases, indicating a progressive maturation of 3D genome organization.

X Chromosome Inactivation Patterns

The paper provides high-resolution views of X chromosome organization, noting changes in compartmental asymmetry and pairing during early germ cell development.

(**article?**)`{Lu2025, year = {2025}, title = {{Single-cell long-read Hi-C, scNanoHi-C2, details 3D genome reorganization in embryonic-stage germ cells}}, author = {Lu, Jiawei and Li, Wen and Tang, Fuchou}, journal = {Nature Structural & Molecular Biology}, doi = {10.1038/s41594-025-01604-7}, issn = {1545-9985}, abstract = {{This study applies scNanoHi-C2 to reveal high-resolution single-cell 3D chromatin structure changes during germline development. Transposable elements and repeat families were differentially enriched in A/B compartments, impacting developmental regulation.}}, pages = {}, number = {}, volume = {}, pmid = {}, pmcid = {}, keywords = {}, local-url = {https://www.nature.com/articles/s41594-025-01604-7} }`

10.0.3 3D genome remodeling and homologous pairing during meiotic prophase

- Simple and LINE repeats enriched in B compartments, involved in meiotic structural integrity.

Key Findings Summary This study integrates Hi-C chromatin conformation capture with super-resolution microscopy to uncover how 3D genome architecture is remodeled during meiotic prophase I in both male and female germlines (spermatogenesis and oogenesis) in mice.

Homologous Chromosome Pairing Chromosomes exhibit precise and progressive pairing during prophase I.

The study captures early pairing events in spermatocytes and oocytes, using high-resolution contact maps.

Contact frequency increases significantly in homologous regions during zygotene and pachytene stages.

2. **Chromatin Compartment Remodeling** A/B compartments are weakened or restructured during prophase.

Specific B compartments, enriched in LINE elements and heterochromatin, show elevated pairing dynamics.

Suggests B compartment flexibility supports pairing more than A.

3. **Role of Repeat Elements** Repeat-rich regions, especially LINEs and LTRs, are preferentially localized to B compartments and correlate with stronger interhomolog interactions.

Repeats may play a structural role in mediating nuclear architecture during meiosis.

4. **Sex Differences in Chromosome Pairing** The authors find distinct dynamics between male and female germ cells, including:

Differences in compartment switching speed.

Asynchronous timing in homolog pairing progression.

Oocytes retain more stable open chromatin than spermatocytes during mid-prophase.

(**article?**)`{He2023, year = {2023}, title = {{3D genome remodeling and homologous pairing during meiotic prophase of mouse oogenesis and spermatogenesis}}, author = {He, Jianrong and Yan, Ao and Chen, Bohan and Huang, Jin and Kee, Kehkooi}, journal = {Developmental Cell}, doi = {10.1016/j.devcel.2023.07.004}, issn = {1534-5807}, abstract = {{By applying Hi-C and super-resolution microscopy, this study tracks chromatin compartment transitions and homolog pairing in meiosis. Repeat-enriched B compartments show increased pairing dynamics and regulate meiotic architecture.}}, pages = {}, number = {}, volume = {}, pmid = {}, pmcid = {}, keywords = {}, local-url = {https://www.cell.com/developmental-cell/abstract/S1534-5807(23)00553-1} }`

10.0.4 DICER regulates repeat RNA and chromosome segregation (maybe???)

- Satellite repeats involved in chromatin architecture, A/B compartment boundary insulation via RNA-based silencing pathways.

(**article?**) {He2023, year = {2023}, title = {{3D genome remodeling and homologous pairing during meiotic prophase of mouse oogenesis and spermatogenesis}}, author = {He, Jianrong and Yan, Ao and Chen, Bohan and Huang, Jin and Kee, Kehkooi}, journal = {Developmental Cell}, doi = {10.1016/j.devcel.2023.07.004}, issn = {1534-5807}, abstract = {{By applying Hi-C and super-resolution microscopy, this study tracks chromatin compartment transitions and homolog pairing in meiosis. Repeat-enriched B compartments show increased pairing dynamics and regulate meiotic architecture.}}, pages = {}, number = {}, volume = {}, pmid = {}, pmcid = {}, keywords = {}, local-url = {https://www.cell.com/developmental-cell/abstract/S1534-5807(23)00553-1} }

11 Acknowledgements

11.1 People

I would like to extend my sincere gratitude to my supervisor, Kasper Munch, for his guidance and support throughout this project. His sparring and scientific literacy have been invaluable in refining my ideas and shaping the direction of this work. I am especially thankful for Kasper's enthusiasm, which has been a source of motivation during this process. His expertise and insights were instrumental in setting up the Quarto templates and providing support for the discussion. His constructive feedback on the manuscript has greatly improved its quality. I feel fortunate to have had the opportunity to learn from and work with such a knowledgeable and dedicated mentor.

I would also like to thank alberte peithersen for the help correcting the report

11.2 Compute

All computations were performed on GenomeDK (GDK), an HPC cluster located on Aarhus University, and most of the processing of the data was handled by a gwf workflow (GenomeDK 2023), a workflow manager developed at GDK. Developing the manuscript and all associated writing, editing, and rendering the project was also performed on GDK so work seamlessly with the analysis. I would like to thank GDK and Aarhus University for providing computational resources and support that contributed to these research results. None of this would have been possible on my own laptop from 2013.

Use of Generative Artificial Intelligence As per Aarhus University guidelines (IT n.d.), I hereby disclose my use of Generative Artificial Intelligence (GAI) tools during the preparation of this thesis. All usage was conducted critically and responsibly, ensuring that the final work remained my own and adhered to academic integrity standards. These tools supplemented my expertise and accelerated routine tasks, but all creative, analytical, and scientific contributions are my own.

11.3 Copilot

I used Copilot for code completion and debugging, as well as modifying existing code to reflect updated variables and experimental needs.

11.4 ChatGPT

ChatGPT provided syntax support, to accelerate my learning of new programming languages. it also help me with proof reading.

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